

ETHAN NIXON

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SKILLS

Softwares: Altium , Cadence Allegro, Cadence OrCAD, KiCAD, PSpice, LTspice, Quartus, Sigrity, Hyperlynx
+ **Programming+Protocols:** C++, Python, Verilog, SystemVerilog, I2C, CAN, SPI, UART, Ethernet, PCIe, GPS
Hardware: Oscilloscope, Logic Analyzer, VNA, TDR, Signal Source Analyzer, DC Supply, Multimeter, FPGA

EXPERIENCE

Arista Networks | *Hardware Design Intern*

Santa Clara, CA | **May 2026–August 2026**

- Executed bring-up, validation, and schematic revisions across **3 boards**: an **LED board test system** driven to its FCS; a **24-layer 6kW PDB** with 54V busbar input, PCIe Gen4 switch, FPGA; and a **12-layer out-of-band test board** with 2.5kW power resistors, FPGA, PCIe switch, and over-temp detection.
- Led architecture, preliminary schematic, **18-layer** stackup design, and high-density connector breakout studies for a **CPU management test card** capable of validating 19× **10G Ethernet**, console, and **SMBus** links.
- Owned architecture, stackup, and schematic design for three high-power, high-density **16+ layer** interposer boards bridging critical power paths for **20,000+** pin BGAs on next-generation main switch cards.
- Designed the clocking and FPGA subsystem in **Cadence OrCAD** and contributed to power architecture design, DPM design, and ASIC strapping for a next generation **1.6TbE** near-packaged optics switch.
- Validated power sequencing of **12+ channels** simultaneously for a 18x OSFP Retimer Card, automating rail-rail delay results across trials via a custom Python script.
- Performed jitter measurements, clock characterization, power rail sequencing and validation on next-generation switch boards, leading efforts in root causing analysis on unit failures during **low and high temp EDVT**.

McMaster Mars Rover | *Electrical Lead & Battery Subsystem Lead*

Hamilton, ON | **Sept 2024–Present**

- **Leading & managing 15+** members to create PCBs such as DC-DC converters, RF communications, BMS, and stepper motor controllers, overseeing **schematic & PCB reviews**, board validation, and assembly.
- Developing a custom **6-layer Battery Management System (BMS)** PCB featuring an **STM32**, passive balancing, **charge and discharge isolation**, and current/voltage sensing for **state of charge** calculation.
- Designed and validated a 4-layer **Teensy 4.1**-based communication board integrating **GPS** and **LoRa** modules with **dual RF power domains**, fan controllers, a servo driver, and diagnostic indicators for telemetry and autonomous control operations.
- Assembled and tested **15+ 24 V stepper motor controller boards**, evaluating electrical performance, efficiency, and thermal stability under varying loads.
- Assembled and validated a **6-layer, 4-phase 48 V → 24 V buck converter** with integrated fault indicators, soft-start behavior, custom thermal mitigation, and an **overvoltage protection stage**.

Dentra | *Electrical Design Intern*

Waterloo, ON | **May 2025 – Aug 2025**

- Designed and validated a **LiPo charging circuit & UV LED driver** prototype for wearable dental diagnostic hardware, implementing **input overvoltage protection** and **charge current limiting**.
- Ensured hardware safety and performance by integrating temperature-protection features and validating **current limits, voltage tolerances**, and thermal margins through iterative bench tests.

PROJECTS

- **Electric Scooter Soldering, C++, 3D Printing** **Present**
Designing a custom electric scooter integrating a BLDC motor, electronic speed controller, a tack-welded Li-ion battery pack, BMS, and an **onboard speed/data display**.
- **Autonomous Robot: C++, Arduino, Soldering** **UtraHacks 2026**
Developed an Arduino robot with ultrasonic and colour sensors able to **autonomously** follow colour paths, avoid obstacles and pick up blocks with a servo driven robot arm, programmed in **C++**.
- **Smart Watch Python, Flask, React, ESP32, OpenCV, EasyOCR, C++** **March 2025**
Designed a smart watch using ESP32 camera images relayed to a **Flask** server. The watch reads labels and performs **TTS**, achieving **85% accuracy**.
- **Sumobot Arduino, C++, Top 5** **Sumobot Competition 2024**
Designed an Arduino based Sumobot, programmed in **C++** that is capable of **autonomously** detecting opposing bots and removing them from the ring.

EDUCATION

McMaster University

Bachelors of Computer Engineering Candidate,

Hamilton, ON

Sept 2024 - April 2028